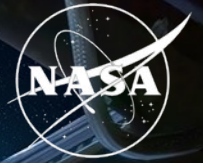


National Aeronautics and
Space Administration



Synthetic transcriptomics for mouse spaceflight

Can generated RNA-seq improve FLT vs GC
analysis within each tissue?

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QUESTION

Small studies and study effects complicate tissue comparisons

Can a generator help rank spaceflight signal without pretending it creates new animals?

NASA OSDR

1,610

biological profiles

75

accessions

835 FLT

775 GC

ARCHS4 mouse

997,515

profiles audited

17,244

selected

20 tissues

- Mission, strain, material and assay differences can resemble a flight effect.
- The generator must preserve tissue and FLT/GC structure without memorizing samples.
- Statistical evidence must still come from observed OSDR profiles.

METHOD

We built a configurable bulk RNA-seq generation pipeline

Data scope, transforms, harmonization, training and conditioning can change without changing the evaluation contract.

Data + scope

OSDR API or ARCHS4
one or many studies
pooled or per tissue

Transform

raw / CPM / TPM
log + scaling
all genes / HVG / L1000

Harmonize

none or study z
ComBat / MBatch
MOBER adapter

Model + train

WGAN-GP or DDIM
OSDR or ARCHS4
pretrain + adapt

Condition

FLT/GC + tissue
study + material
available covariates

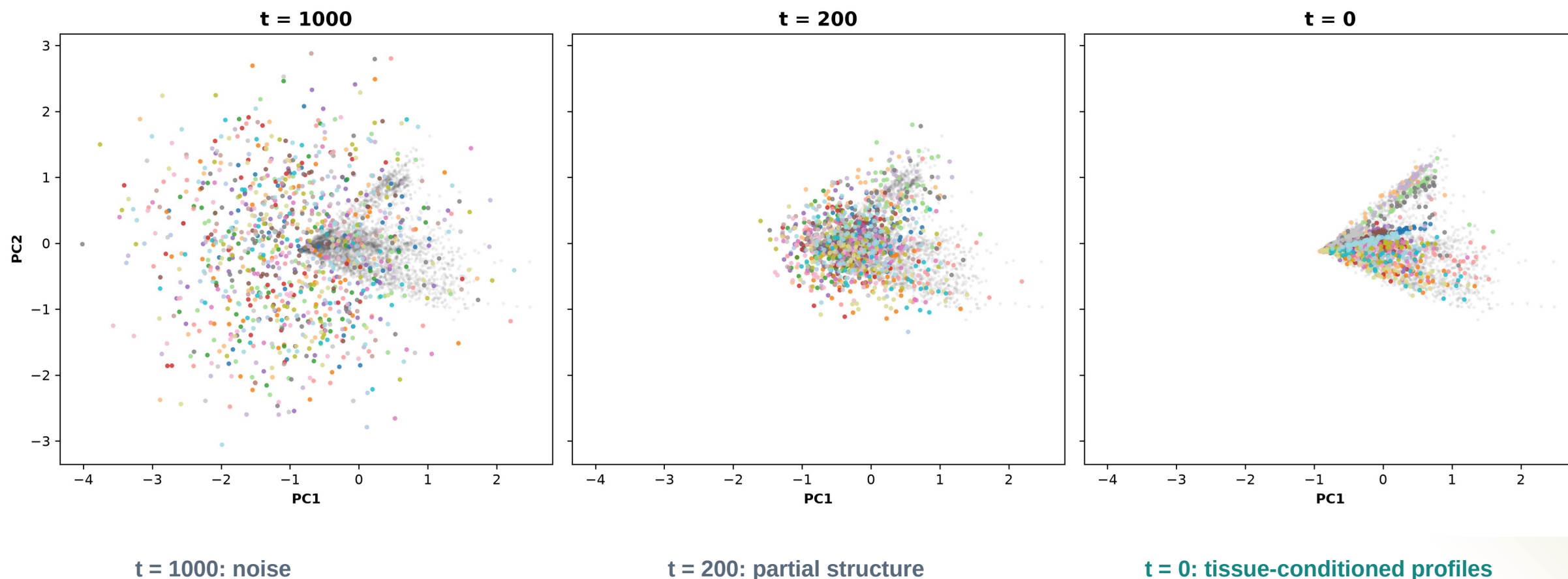
Selected branch **TPM / MaxAbs -> ARCHS4 DDIM -> OSDR adaptation.** Conditions: tissue, FLT/GC, accession and material; no global batch correction.

A 463-row experiment plan and nine matched liver harmonization arms were screened before full training.

DIFFUSION

Diffusion learns tissue structure from noise

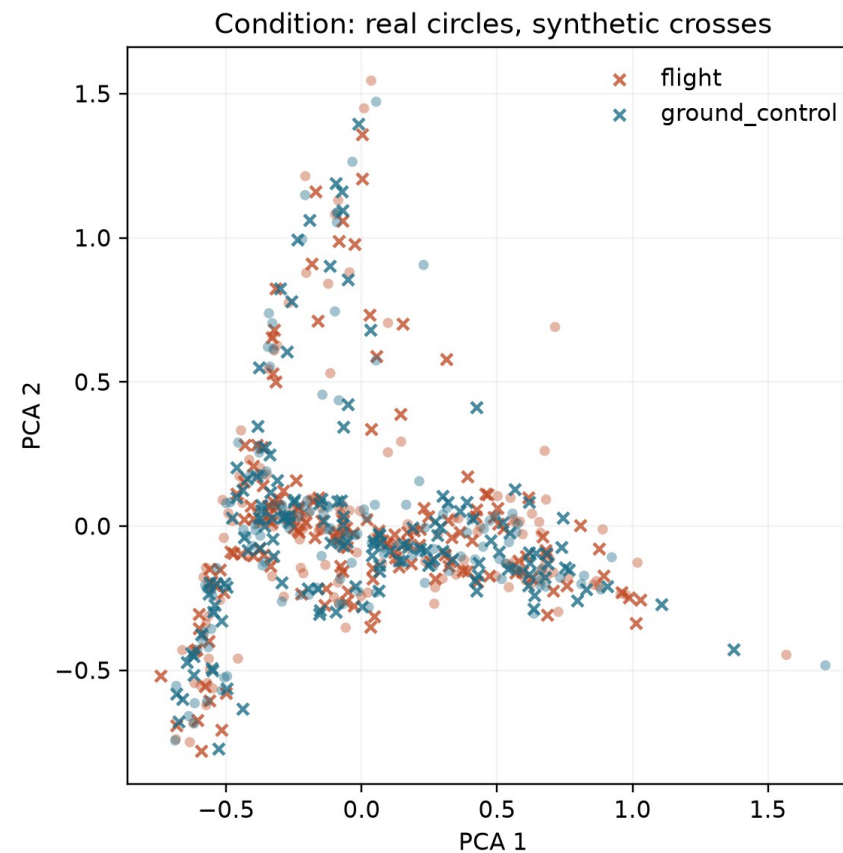
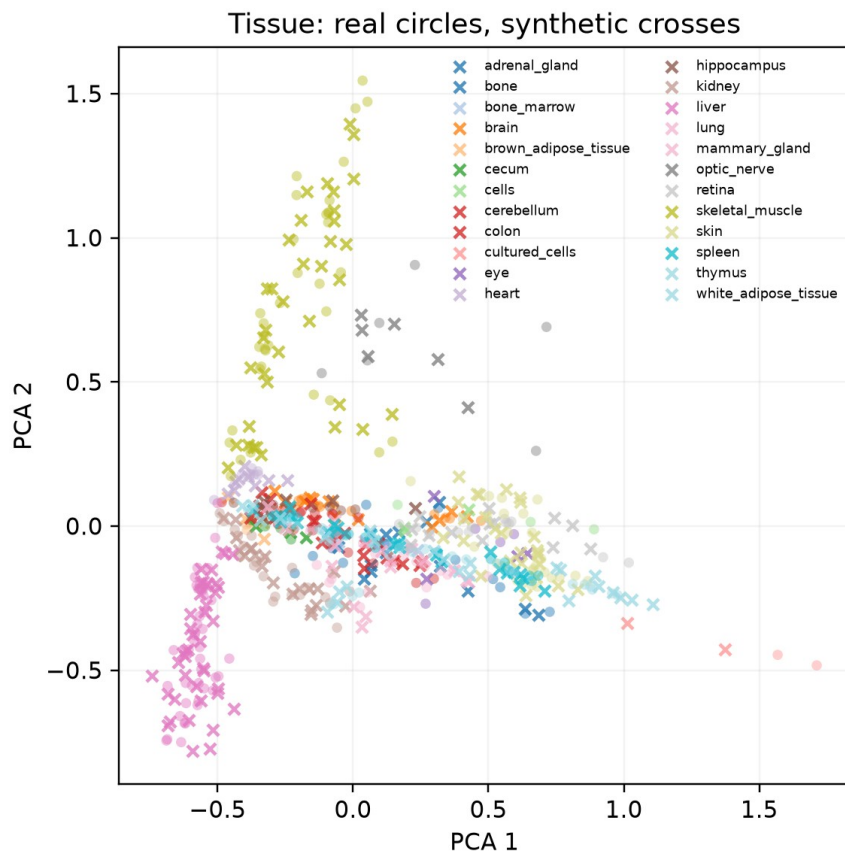
The same PCA axes follow 1,024 generated profiles through reverse diffusion.



DIFFUSION OUTPUT

Generated profiles track the real OSDR PCA manifold

Circles are locked real OSDR profiles; crosses are matched DDIM profiles from generation seed 5020.



Interpretation: tissue structure dominates; FLT and GC are subtler. PCA is descriptive, so quantitative metrics determine validation.

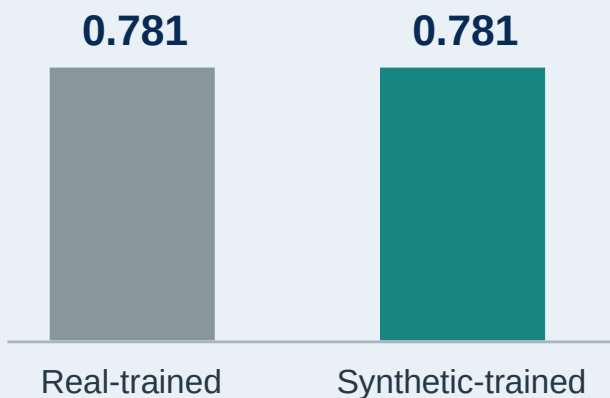
VALIDATION

DDIM matched expression and reduced separability

AA closer to 0.5 means a classifier has less success separating real and generated profiles; lower FD is better.

ARCHS4 tissue probe

Balanced accuracy on held-out real profiles



Metric

WGAN-GP

DDIM

Reading

Correlation

0.976

0.974

similar

F1

0.985

0.997

higher

Adversarial accuracy

0.636

0.475

closer to 0.5

FD / real P95

0.144

0.074

lower

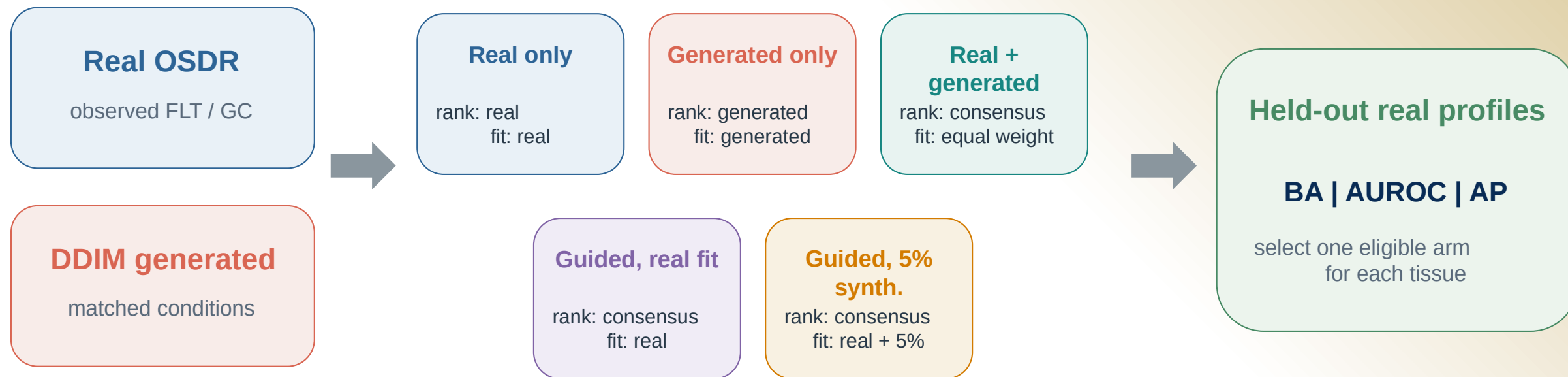
Use DDIM downstream

Lower separability and distributional distance, with comparable correlation and higher F1.

ANALYSIS

Synthetic profiles entered the analysis in five different ways

Every arm was judged on held-out real profiles before synthetic attribution was allowed.



Biology check FLT vs GC effects, random-effects meta-analysis and BH FDR use real OSDR profiles only.

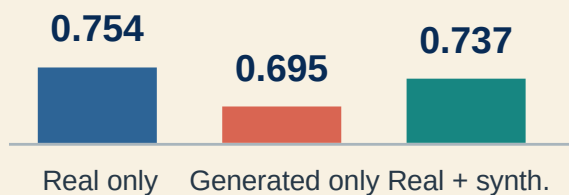
Synthetic profiles never increase animal n.

Pooling tissues hid useful signal

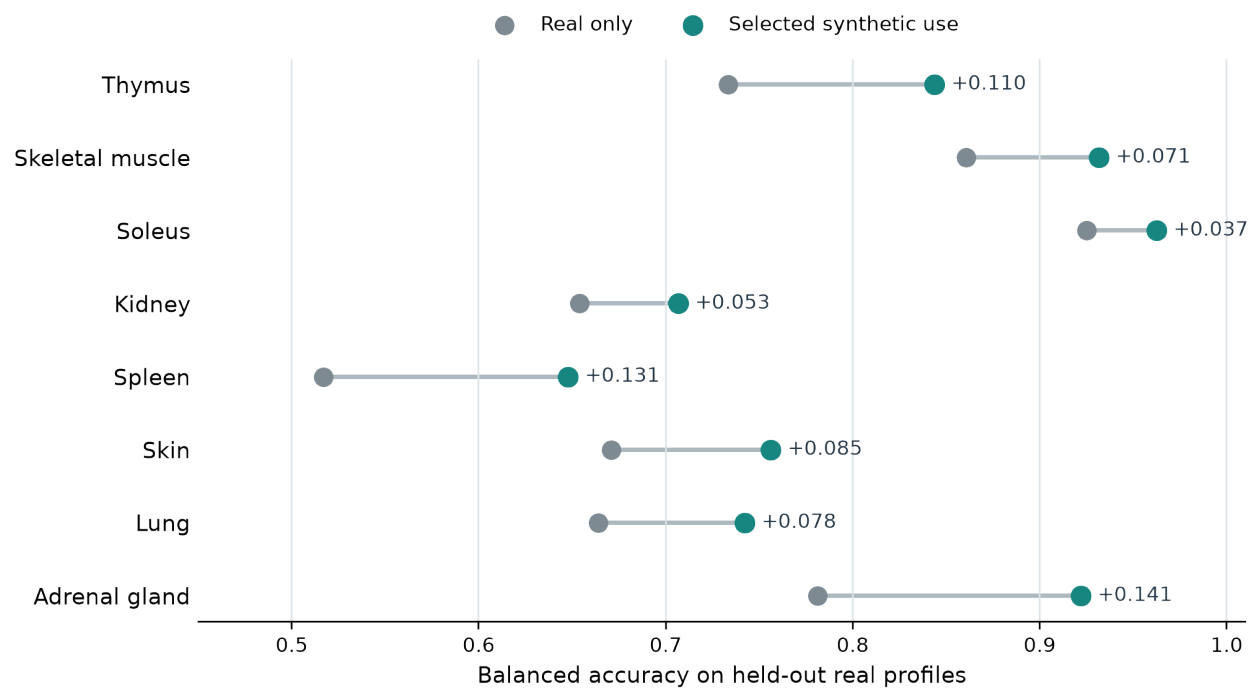
A single FLT/GC classifier did not improve, but tissue-specific synthetic use often did.

Pooled across tissues

Balanced accuracy



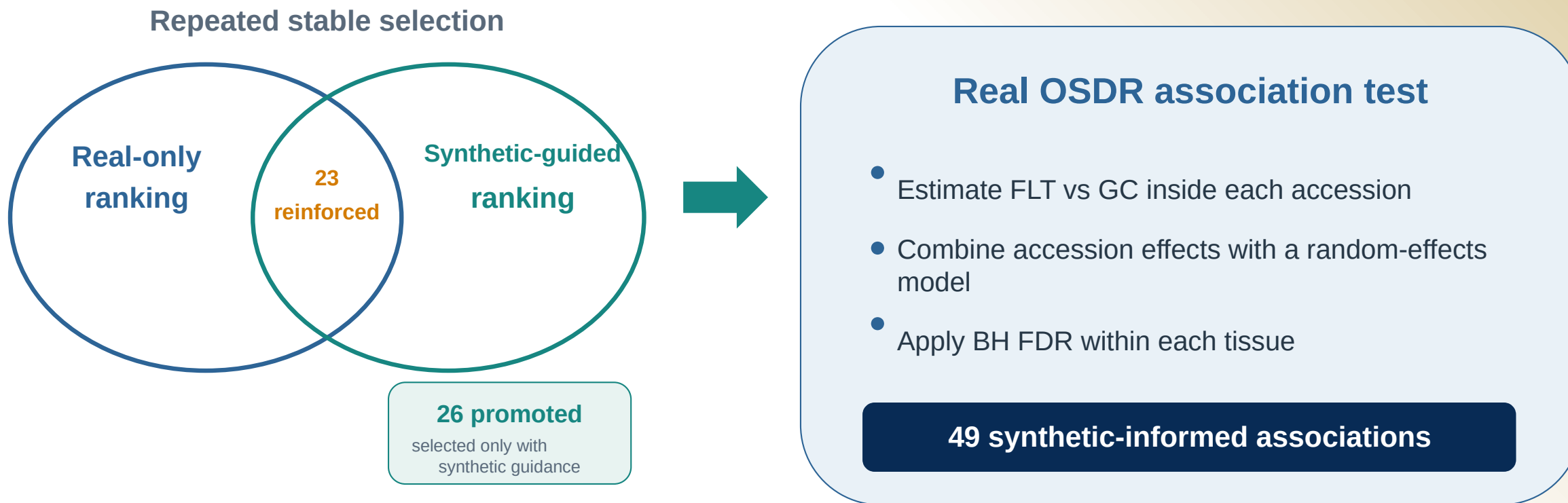
Selected use within each tissue



The useful arm differed by tissue.

Synthetic guidance changed ranking, not statistical evidence

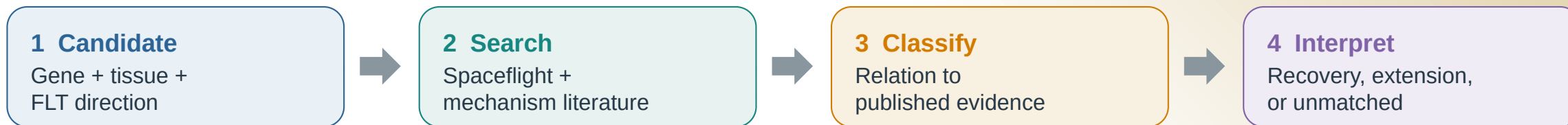
Promoted and reinforced describe repeated feature selection; both require a supporting effect in real OSD data.



These are 49 tissue-gene associations among 459 BH-FDR rows. A gene can appear in more than one tissue.

Annotation separates recovery from hypothesis extension

The label describes the relationship to prior evidence, not whether a gene is biologically plausible.



What the 26 promoted associations showed

- | | | |
|-----------|-----------------------------|--|
| 11 | Aligning | 3 exact matches + 8 same-tissue process matches |
| 13 | Complementary | Related process or mechanism; not direct replication |
| 1 | Ambiguous | Birc5 differed across mission contexts |
| 1 | Literature-unmatched | Psmb8 remains mechanistically plausible |

How to read a result

- Promoted: synthetic guidance changed stable ranking
- BH FDR: the association is present in real OSDR data
- Literature class: direct, related, mixed, or unmatched
- Interpretation: a testable hypothesis, not independent proof

11 align and 13 extend prior biology; only 3 are exact gene-tissue-direction matches.

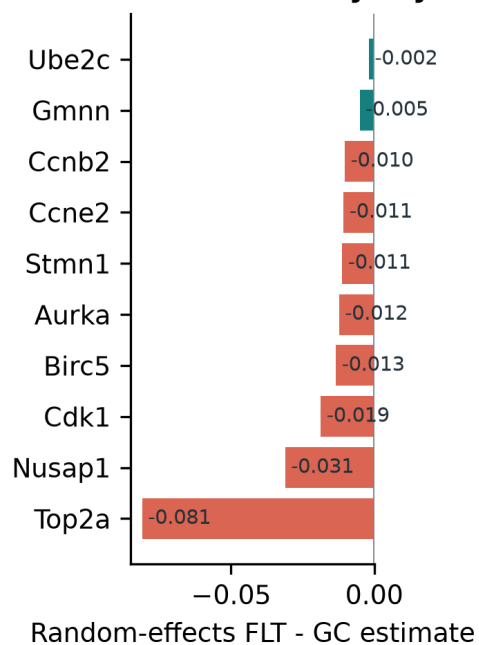
THYMUS

Thymus points to lower proliferative renewal

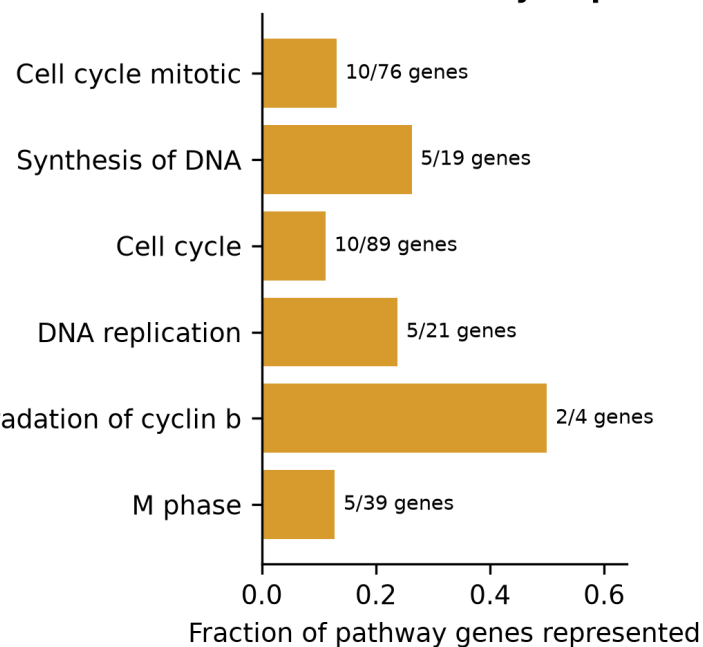
The core panel recovers cell-cycle biology; two flight-higher genes extend the hypothesis.

Synthetic-informed thymus genes converge on a flight-lower mitotic program

A Cross-study thymus effects



B Shared cell-cycle processes

**16**

synthetic-informed
BH-FDR associations

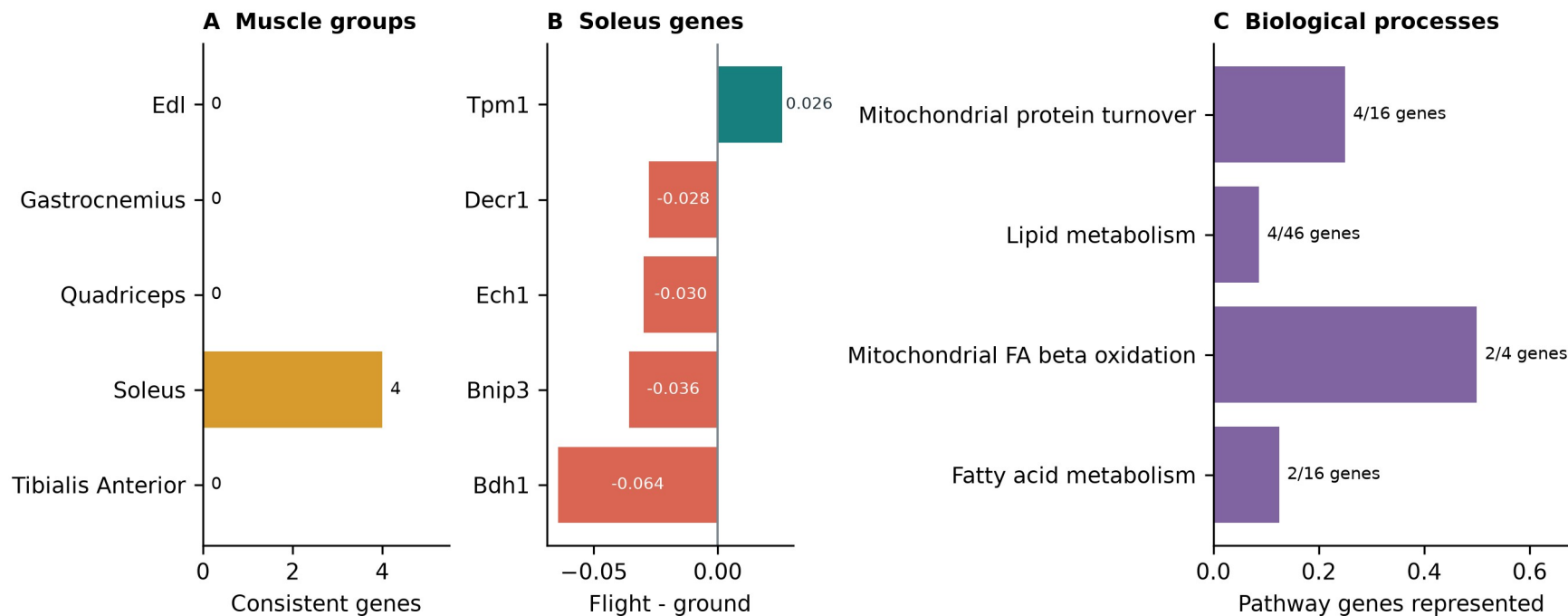
13 promoted | 3 reinforced

- Lower Nusap1, Stmn1, Birc5, Cdk1, Top2a and related genes
- Cell cycle, DNA replication, APC/C and G2/M processes
- Higher Hsd17b11 and Etv1 suggest lipid or T-cell-state shifts
- Bulk RNA-seq cannot separate cell loss from lower transcription

Soleus reinforces a mitochondrial and lipid program

This result strengthens an existing real-data pattern rather than promoting a new soleus gene.

Anatomical separation reveals a soleus-specific oxidative-metabolism hypothesis



0.925 -> 0.963

balanced accuracy

5 reinforced | 0 promoted

- Lower Bdh1, Ech1, Bnip3 and Decr1
- Higher Tpm1
- Mitochondrial turnover and fatty-acid metabolism
- Compatible with unloading and contractile remodeling

OTHER TISSUES

Kidney adds a focused signal; other tissues are exploratory

The strongest process-level stories were thymus and soleus, but several tissues produced narrower candidates.

Kidney	Inpp4b promoted, Slc37a4 reinforced; both higher in flight	Focused secondary
Spleen	Rai14, Ptprk and Myl9 promoted; Loxl1 reinforced; no Reactome enrichment	Exploratory
Skin + adrenal	Plscr1 higher in skin; adrenal Psmb8 is plausible but literature-unmatched	Exploratory
Lung + retina	Predictive gains without a synthetic-informed BH-FDR gene	Prediction only
Liver + EDL + quadriceps	Real-only arm retained	No synthetic claim

A plausible mechanism is not independent validation; these candidates still require targeted experiments.

TAKEAWAYS

Synthetic data helped most as a tissue-specific prior

The generator created useful structure, not new biological replication.

1 Generate

Conditional DDIM produced high-fidelity profiles with near-chance real-versus-synthetic separation.

2 Use

Pooled augmentation failed. Tissue-specific ranking and low-weight training were more useful.

3 Interpret

Thymus and soleus were clearest. Literature review separated recovery from complementary hypotheses.

Next test

Use independent samples and cell-resolved assays to test thymus proliferation, soleus metabolism, kidney signaling and adrenal Psmb8.

Generated profiles never enter the biological sample count or the BH-FDR test.

Thank you

Questions?

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Mentor

James Casaletto

Program

NASA Space Life Sciences Training Program

Data

NASA OSDR | ARCHS4 | Reactome

Compute

NASA Ames Research Center

Code and
manuscript

github.com/jasont314/nasa-mouse

All biological associations were tested in observed OSDR profiles.